

Cu Nanoclusters with Aggregation Induced Emission Enhancement

Xiaofang Jia, Jing Li, and Erkang Wang*

A facile and versatile method for preparing water-soluble, stable, luminescent Cu nanoclusters (NCs) via the process of size-focusing etching from nonluminescent nanocrystals is presented. Using glutathione as a model ligand, the smallest cluster, Cu₂, is selectively synthesized to form a nearly monodisperse product, eliminating the need for tedious size fractionation. Evolution of photoluminescence and absorption spectra reveal that the formation of stable cluster species occurs through surface etching. Intriguingly, the as-prepared CuNCs exhibit an aggregation-induced emission enhancement effect. The CuNCs emit a faint light when dispersed in aqueous solution, but generate a striking fluorescence intensity enhancement upon aggregation. Armed with these attractive properties, the emissive CuNCs are expected to open new opportunities for the construction of light-emitting diodes, chemosensors, and bioimaging systems.

1. Introduction

Metal nanoclusters (NCs) have received much attention because their unique physical, electrical, and optical properties are very promising for their application in catalysis, chemical sensors, electronic devices, and biological imaging.^[1] Typically, metal NCs, consisting of several to tens of metal atoms, bridge the gap between metal nanoparticles and small molecules.^[2] With a size approaching the Fermi wavelength of electrons, metal NCs display discrete energy levels, leading to the observation of interesting molecule-like properties such as strong fluorescence. Compared with common fluorophores such as organic dyes and semiconductor quantum dots, metal NCs are appealing alternatives due to their ultrafine size, excellent photostability, large Stoke's shift,

and low environmental hazard.^[3] Developing a simple and general synthetic route is highly desirable for both fundamental studies and practical applications of this new type of material. Recently, fluorescent AuNCs and AgNCs have been extensively studied because of their chemical stability and facile synthetic procedure.^[4] Compared with noble metals Au and Ag, the metal Cu is relatively abundant, inexpensive, and readily available from commercial sources. Due to the synthetic difficulty in controlling ultrafine size and the susceptibility to oxidation on exposure to air, it is still a challenging task to prepare fluorescent CuNCs. Unlike the cases of Au and Ag, there are only limited studies focusing on the synthesis of fluorescent CuNCs.^[5]

The conventional synthetic procedure based on nucleation of zerovalent Cu atoms in the presence of thiols suffers from the production of a mixture of different cluster sizes.^[5a] Preparation of atomically monodisperse CuNCs is of primary importance, allowing in-depth studies of their attractive properties and promoting their further practical application. To achieve this goal, careful and complicated post-synthetic treatments are often involved to purify CuNCs, increasing the overall time and synthetic cost. Thus, it is highly desirable to develop synthetic strategies that allow the size-selective synthesis of monodisperse CuNCs, eliminating the need for size fractionation. The thiol-mediated etching methodology, referred to as a "size-focusing" process, opens up the

X. F. Jia, J. Li, Prof. E. K. Wang
State Key Laboratory of Electroanalytical Chemistry
Changchun Institute of Applied Chemistry
Chinese Academy of Sciences
Changchun, Jilin, 130022, China
E-mail: ekwang@ciac.jl.cn

X. F. Jia
Graduate School of the Chinese Academy of Sciences
Beijing, 100039, China



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possibility of selective synthesis of a number of atomically monodisperse ultrasmall AuNCs and AgNCs, such as Au₂,^[6] Au₈,^[7] Au₂₅,^[4c] Au₃₈,^[8] Au₁₄₄,^[9] Ag₈,^[10] and so on. Due to the peculiar stability of certain-sized metal NCs, the most robust ones can survive during the size-focusing etching process, much like the natural law “survival of the fittest”.^[11] In this work, we converted the nonluminescent Cu nanocrystals to luminescent CuNCs via the size-focusing etching process in the presence of excess ligands. Using glutathione (GSH) as a model ligand, the smallest cluster Cu₂ was selectively synthesized to form a nearly monodisperse product. Evolution of photoluminescence (PL) and absorption spectra revealed that the formation of stable cluster species occurred through the surface-etching route. Intriguingly, the as-prepared CuNCs exhibited an aggregation-induced emission enhancement (AIEE) effect. The CuNCs emitted a very faint light when dispersed in aqueous solution, but gave rise to a striking fluorescence intensity enhancement upon aggregation. The general utility of the procedure was further demonstrated by the synthesis of fluorescent CuNCs protected by different types of thiol ligands.

2. Results and Discussion

CuNCs are prepared via a two-step process: the synthesis of small Cu nanocrystals and subsequent conversion into CuNCs. Cu nanocrystals were facilely prepared by reduction of Cu²⁺ ions using ascorbic acid (AA) as both reducing

agent and capping agent at room temperature.^[12] Typically, Cu(NO₃)₂ aqueous solution (0.1 M, 1 mL) was added dropwise into the AA solution (0.1 M, 10 mL) under vigorous stirring. The reaction mixture was left to stir for 1 h at room temperature. Then a natural tripeptide GSH was adopted as a model etching agent for a ligand exchange reaction with the colloidal Cu nanocrystals. The as-prepared Cu nanocrystal solution (0.3 mL) was added to the GSH aqueous solution (0.05 M, 3 mL) and kept stirring for 5 h at room temperature. Transmission electron microscopy (TEM) showed the synthesized Cu nanocrystals to have a high monodispersity with a mean diameter of 4.2 ± 1.1 nm (Figure 1A,B). Upon careful measurement from the high-resolution TEM (HRTEM) images of Cu nanocrystals (Figure 1A inset), fringes with an interplanar spacing of 0.207 nm were identified, which is attributed to the (111) diffraction plane of face-centered Cu (JCPDS 89-2838). After the etching process, TEM revealed that large nanocrystals decomposed and clusters appeared with average diameters of 1.3 ± 0.4 nm (Figure 1C,D).

Mass spectroscopy has turned out to be a very powerful method for the investigation of NCs. The chemical composition of the CuNCs was analyzed using matrix-assisted laser desorption ionization time-of-flight mass spectroscopy (MALDI-TOF MS). The spectrum was collected in negative-ion mode using the matrix sinapinic acid. As shown in the Figure 2A, a major peak at m/z 739.07 corresponds well with the molecular formula [Cu₂L₂-H]⁻, (theoretical molecular weight: 738.99, deviation 0.08, where L = C₁₀H₁₆O₆N₃S). This assignment is further supported by the perfect match of the simulated and experimental isotopic distributions (Figure 2A inset). The lower mass peak at m/z 675.08 is ascribed to CuL₂⁻ (Figure S1). No higher mass peak was observed even using other matrices, such as 2,5-dihydroxybenzoic acid and α -cyano-4-hydroxycinnamic acid, demonstrating the monodispersity of the as-prepared CuNCs. To our knowledge, these are the smallest CuNCs obtained by solution-phase chemical synthesis, though it could not be precluded that the larger CuNCs observed in the TEM images did not ionize easily and went undetected in the MS sample, and only the fragments like CuL₂⁻ and [Cu₂L₂-H]⁻ were detected. FTIR spectra were used to get further insight into the chemical and surface properties of the as-prepared CuNCs. As shown in Figure 2B, the FTIR spectrum of CuNCs shows the characteristics of the ligand GSH, although broad spectral features are found as compared to the pure ligand. The peak at 2524 cm⁻¹ that corresponds to the S-H stretching vibration mode disappeared in the CuNCs, indicating that the GSH molecules anchor on the CuNC surface through Cu-S bonding. No feature of AA was observed, which confirmed that the ligand exchange was

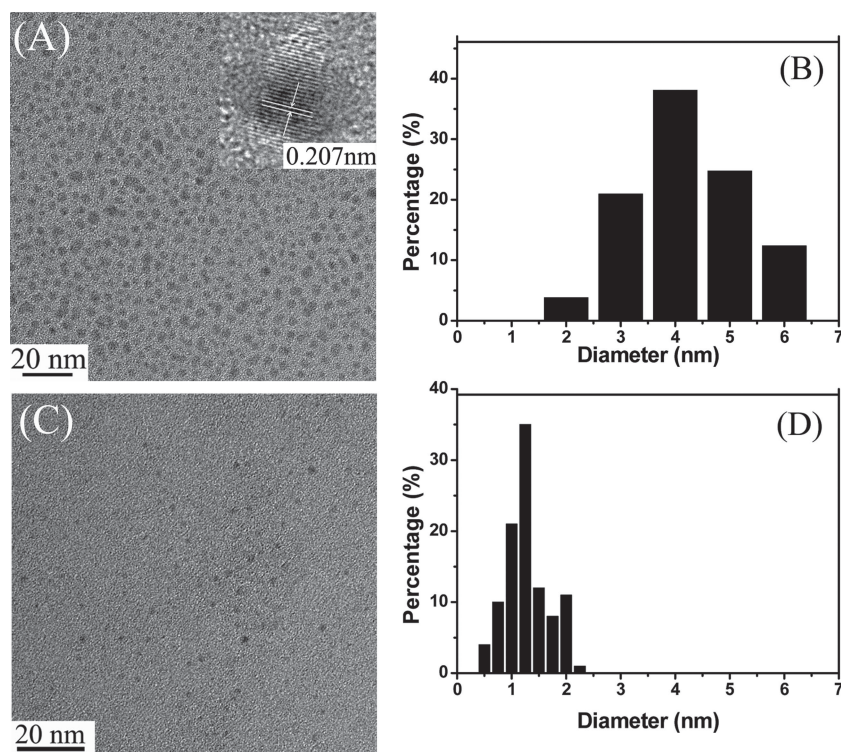


Figure 1. The representative TEM image (A) before and (C) after etching the Cu nanocrystals. Inset: HRTEM image of Cu nanocrystals showing lattice fringes consistent with the (111) diffraction plane of face-centered Cu. The size distribution (B) before and (D) after etching the Cu nanocrystals.

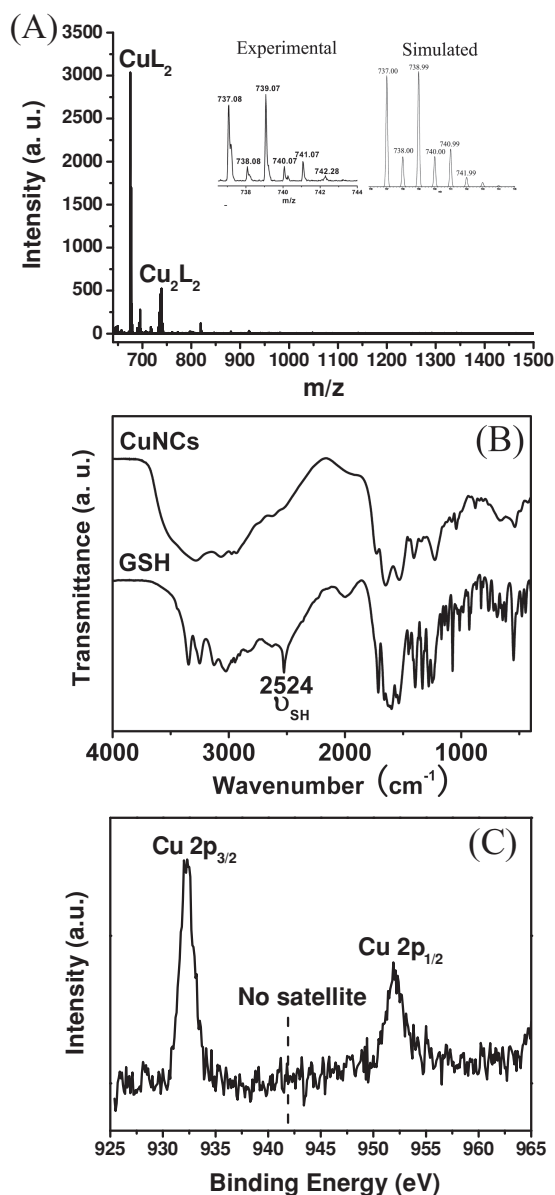


Figure 2. (A) Negative mode MALDI-TOF MS of CuNCs (where L stands for $C_{10}H_{16}O_6N_3S$); Inset shows the experimental and simulated isotopic pattern of $[Cu_2L_2-H]^-$; (B) FTIR spectra of pure GSH and GSH-capped CuNCs; (C) XPS of Cu 2p electrons in CuNCs; no satellite peak confirms the absence of Cu^{2+} .

complete due to the stronger affinity of Cu–S than of Cu–O. X-ray photoelectron spectroscopy (XPS) analysis was carried out to determine the oxidation state of Cu in the samples. An XPS survey spectrum shows that samples are composed of all the expected elements C, O, N, S, and Cu (Figure S2). No characteristic satellite peak around 942 eV implies the absence of Cu^{2+} in the CuNCs (Figure 2C).^[13] Two intense peaks at 951.9 and 932.3 eV are assigned to Cu $2p_{1/2}$ and $2p_{3/2}$, which are attributed to Cu(0) and consistent with the previous report.^[14] However, it is worth mentioning that the $2p_{3/2}$ binding energy of Cu(0) is very close to that of Cu(I) species.^[5a] Also, due to charge transfer in the Cu–S bonds, the valence state of the obtained CuNCs most likely lies between 0 and +1.

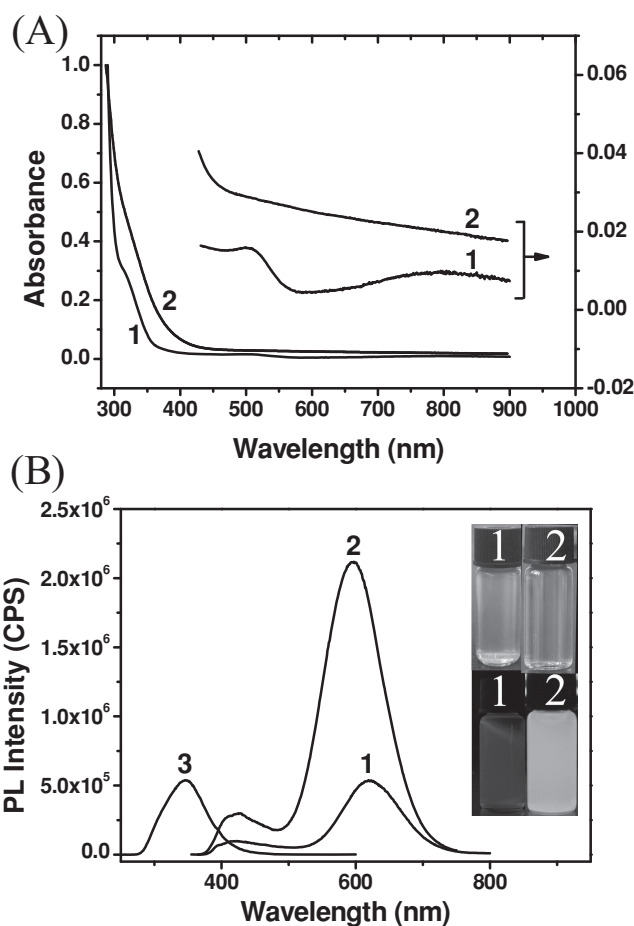


Figure 3. (A) Absorption spectra of the as-prepared Cu nanocrystals (curve 1) and CuNCs (curve 2); (B) excitation spectrum ($\lambda_{em} = 620$ nm, curve 3) and emission spectra ($\lambda_{ex} = 345$ nm) of CuNCs in aqueous solution (curve 1) and 80 vol% ethanol solution (curve 2); Inset: photographs of the CuNCs in aqueous solution (1) and 80 vol% ethanol solution (2) under room light (top) and 365 nm irradiation (bottom); the concentration of CuNCs in aqueous solution (0.8 mM in terms of the concentration of Cu) is 10-fold of that in 80 vol% ethanol solution.

As depicted in **Figure 3A**, absorption spectra showed that the characteristic absorption peak at 507 nm arising from the surface plasmonic resonance of Cu nanocrystals disappeared after etching, and was replaced with a monotonically exponentially increasing spectrum toward shorter wavelengths, indicating the formation of CuNCs.^[15] The as-prepared CuNC aqueous solution displayed a faint red color under UV light (Figure 3B inset). In contrast, upon addition of ethanol or methanol, the CuNCs were easily precipitated out from the supernatant and emitted an intense orange light (Figure S3 and Figure 3B inset), indicating that the as-synthesized CuNCs exhibited an AIEE effect. The aggregation of CuNCs in the aqueous solution by the addition of ethanol is reversible; addition of water to aggregated CuNCs resulted in the redispersion of the clusters to re-form a clear solution. This procedure is applied to separate the clusters from the solution. **As shown in Figure 3B, upon excitation at 345 nm, the CuNCs in aqueous solution displayed one weak emission peak at 620 nm. Upon addition of ethanol, a strong emission peak at 600 nm and a minor emission peak at 426 nm**

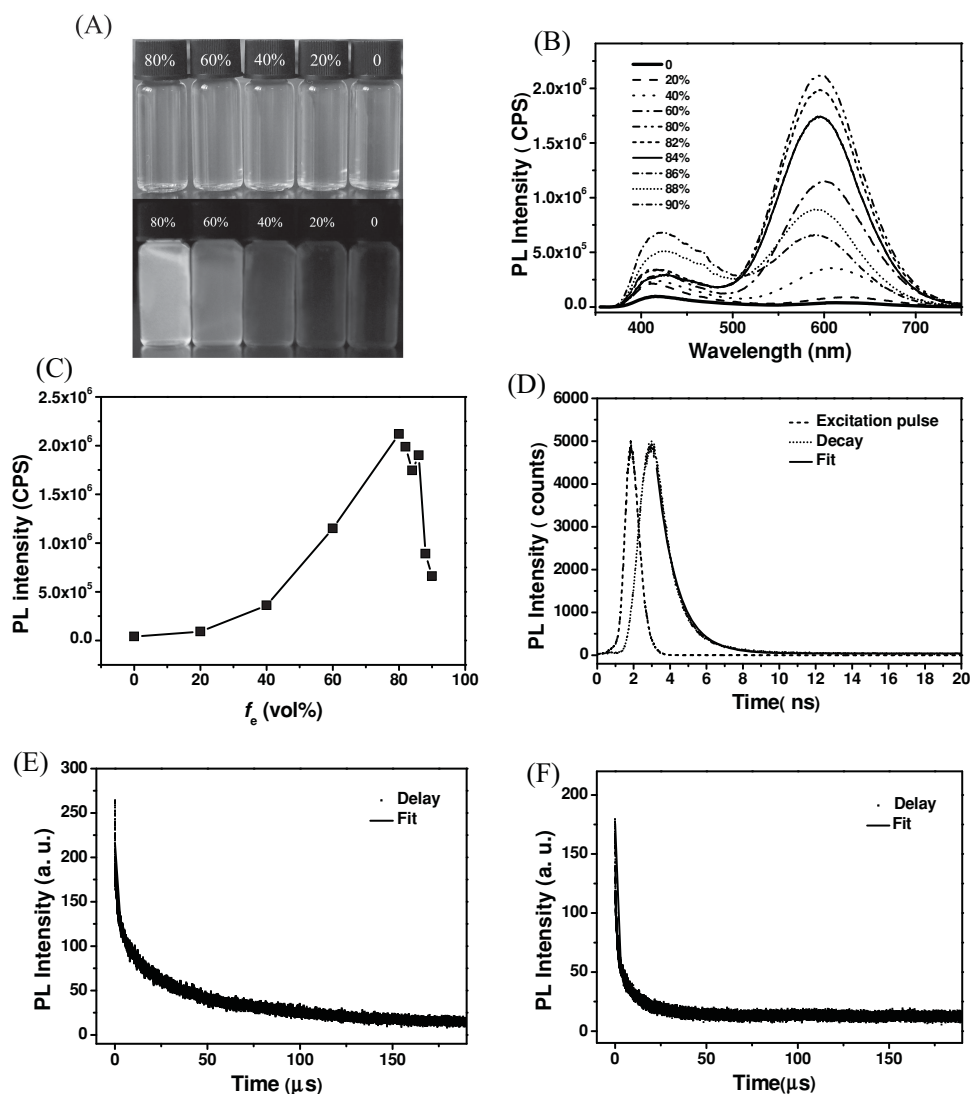


Figure 4. (A) Photographs of the CuNCs in ethanol/water mixtures with different f_e under room light (top) and 365 nm irradiation (bottom); (B) emission spectra of the CuNCs in the ethanol/water mixtures with different f_e ; (C) plots of PL intensity of CuNCs versus f_e ; (D) PL decay profile ($\lambda_{em} = 425$ nm) of CuNCs in the 80 vol% ethanol at the excitation of 377 nm; (E) PL decay profile ($\lambda_{em} = 600$ nm) of the CuNCs in the 80 vol% ethanol at the excitation of 355 nm; (F) PL decay profile ($\lambda_{em} = 620$ nm) of the CuNCs in aqueous solution at the excitation of 355 nm. Fitted curves were overlaid on the experimental data.

were observed, accompanied by the striking enhanced PL intensity. When dissolved in aqueous solution, the quantum yield (QY) of CuNCs was as low as 0.45% at room temperature. In sharp contrast, when CuNCs were aggregated upon addition of ethanol (80 vol% fraction), the QY of CuNCs rose to 6.6%, which was 14.7-fold higher than that in pure aqueous solution. The powder of the CuNCs is stable for several months by storing it in a refrigerator without the protection of an inert gas. The improved stability facilitates their usability in catalysis and other sensing applications.

To get a clear picture about the AIEE effect, we further examined the PL behavior of the CuNCs in ethanol/water mixtures with different volumetric fractions of ethanol (f_e , vol%). As illustrated in **Figure 4A**, the CuNCs aggregates generated with increasing f_e , and simultaneously emissive light under 365 nm irradiation was gradually intensified. As depicted in

Figure 4B,C, the PL intensity at 600 nm increased progressively with f_e and reached the maximum at 80 vol%. When f_e was further increased to 88 vol%, the PL intensity at 600 nm dropped, concomitant with the appearance of obvious precipitates. Instead, the PL intensity at 425 nm continued to increase, probably due to the solvent polar effect. In the case, when that the CuNCs were phase-transferred to toluene, the PL intensity at 425 nm was also comparably high (**Figure S4**).

To get insight into the PL mechanism, the PL lifetime of the CuNCs at different emission wavelengths was measured. **Figure 4D,E** show the PL decay of CuNCs in the 80 vol% ethanol. The corresponding fitted curve was calculated with an exponential decay based on a non-linear least-squares method (**Figure 4D**). The PL lifetime of the emission at 425 nm is 1.13 ns, which is in the scope of PL decays of many fluorescent zero-value metal NCs and attributed to emission

Table 1. Lifetimes of GSH-mediated CuNCs in different solvent.

solvent	$\lambda_{em}/\lambda_{ex}$ [nm]	Lifetime [μ s]			
		τ_1 (A_1)	τ_2 (A_2)	τ_3 (A_3)	τ_{ave}^a
80 vol% ethanol	600/355	0.74 (1.1%)	9.22 (12.1%)	60.83 (86.7%)	53.86
water	620/355	0.17 (1.7%)	1.71 (15.2%)	12.78 (83.1%)	10.88

^a) The average lifetime was calculated according to $\tau_{ave} = \Sigma A_i \tau_i$.

from singlet excited states.^[3,5d,16] As listed in the **Table 1**, the PL decay at 600 nm was described with a three-exponential fitting, giving an average lifetime of 53.86 μ s (Figure 4E). The long excited-state lifetime and the large Stoke's shift (230 nm) suggest that the emission at 600 nm could be assigned to triplet excited states, which is consistent with the observed ligand–metal charge transfer behavior.^[5e,17] Similar features were observed in the time-decay behavior of CuNCs in aqueous solution, with a relative short average lifetime of 10.88 μ s (Figure 4F). In the aqueous solution, the CuNCs are dissolved as isolated species and little restriction is imposed on their molecular vibrational, rotational, and torsional movements. The relatively short lifetime and low QY of CuNCs in aqueous solution implies that molecular motion can serve as a nonradiative deactivation (decay) process for the excited state.^[18] In the aggregates, however, the intramolecular motions are restricted, due to the physical constraint which blocks the nonradiative path and activates radiative decay. Thus, in the aggregates, the PL lifetime is longer, concomitant with the enhanced emission efficiency. Nevertheless, we surmised that the restriction of

intramolecular motion in the aggregates could be a cause for the AIEE effect.^[19]

Two possible routes can be considered for the “top-down” size-focusing etching process.^[20] One possible route would be “surface etching”: Cu atoms are removed from the surface of the nanocrystals by excess GSH as a Cu(I)–GSH complex, and then the Cu(I)–GSH complex tends to form Cu₂ clusters via Cu⁺...Cu⁺ cuprophilic interaction.^[21] An alternative would be “size reduction”: GSH molecules etch the surface Cu atoms of the Cu nanocrystals, leading to a reduction in size of the nanocrystals in steps, and finally Cu₂ clusters are produced, accompanied by the resultant Cu(I) thiolate deposit. As illustrated in **Figure 5A**, in our reaction system, the luminescent CuNCs appeared immediately after addition of GSH. However, nearly no change was perceivable for the characteristic absorption peak at 507 nm arising from the surface plasmonic resonance of Cu nanocrystals (Figure 5B), indicating that the size was retained. Furthermore, a typical TEM image of Cu nanocrystals after etching for 3 min is shown in Figure 5C. The average particle diameter was found to be 3.3 ± 0.7 nm, as indicated in the histogram (Figure 5D). Restricted by the resolution of TEM, the very small cluster Cu₂ could not be observed. Then the CuNCs were further precipitated and separated from the larger Cu nanocrystals after etching for 3 min. As shown in Figure 5E, MALDI-TOF MS was applied to characterize the purified CuNCs. The series of peaks are ascribed to CuL₂[−] and [Cu₂L₂−H][−], respectively. Detailed peak assignments are displayed in Figure S5. Thus it is reasonable to assume that the luminescent CuNCs were produced via the route of surface etching but not successive size reduction. As the reaction proceeded, some white aggregate was yielded. The absorption peak at 507 nm disappeared,

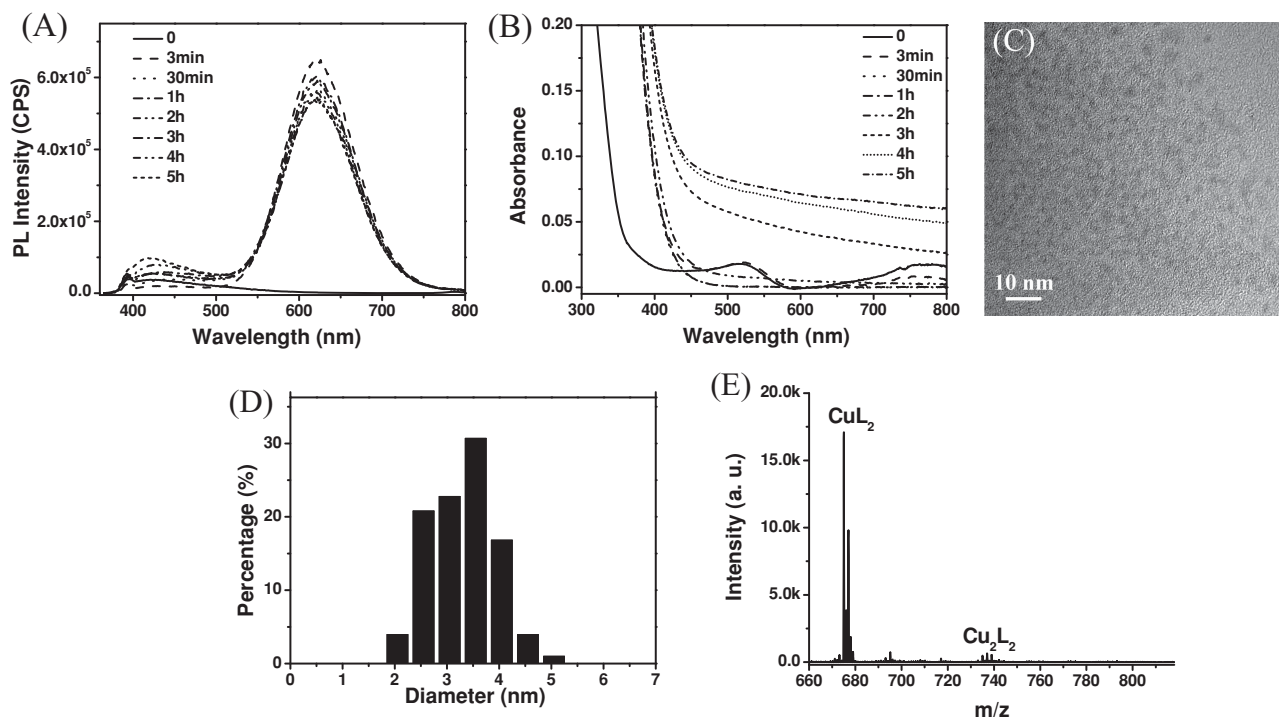


Figure 5. Time-resolved evolution of (A) PL spectra and (B) optical absorption spectra of CuNCs; (C) TEM image and (D) size distribution histogram of Cu nanocrystals after etching for 3 min; (E) negative mode MALDI-TOF MS of CuNCs after etching for 3 min.

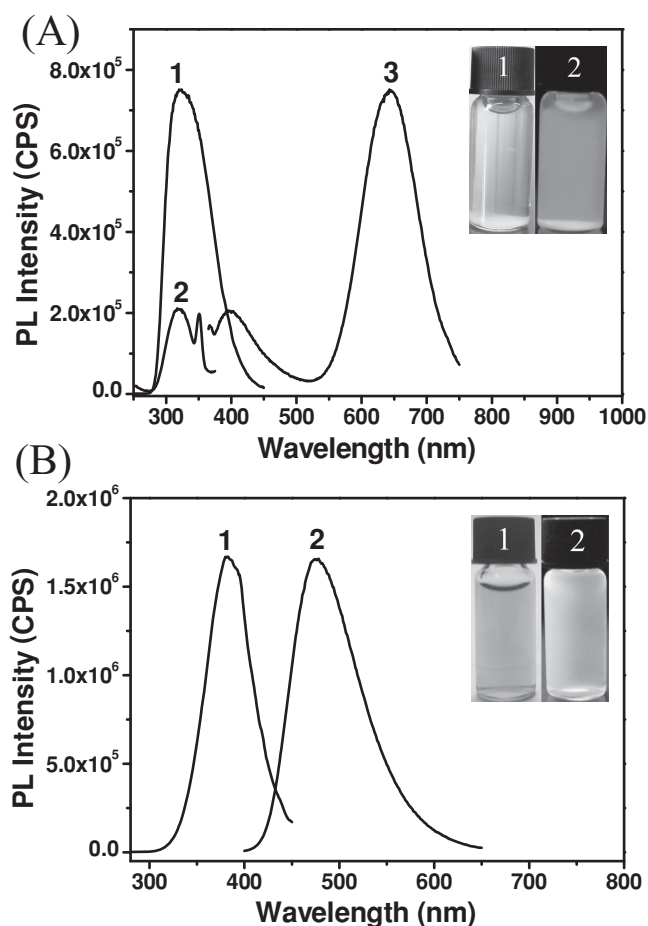


Figure 6. (A) Excitation spectrum ($\lambda_{em} = 645$ nm, curve 1; $\lambda_{em} = 398$ nm, curve 2) and emission spectra ($\lambda_{ex} = 322$ nm, curve 3) of Pen mediated CuNCs in aqueous solution. Inset: photographs of the CuNCs under room light (1) and 365 nm irradiation (2); (B) excitation spectrum ($\lambda_{em} = 478$ nm, curve 1) and emission spectra ($\lambda_{ex} = 382$ nm, curve 2) of Cys mediated CuNCs. Inset: photographs of the CuNCs under room light (1) and 365 nm irradiation (2).

and was replaced with a monotonically increasing spectrum roughly exponentially toward shorter wavelengths, demonstrating that large nanocrystals were decomposed. After etching for 5 h, the TEM image in the Figure 1C shows that the size of Cu nanocrystals decreased to 1.3 ± 0.4 nm. On the other hand, the PL intensity changed little, indicating these larger clusters were not luminophors. This further demonstrates that luminescent CuNCs were not generated through the “size reduction” route. Because Cu is more reactive than the noble metal, the decomposition kinetics were much quicker than those of previously reported noble metal NCs by the “top-down” approach, which required a much longer time for reaction completion.^[10,16,22]

To effectively prepare the CuNCs, the concentration of GSH was critical (Figure S6). In a low concentration of GSH (10 mM), only a white precipitate Cu(I)–GSH was obtained after etching. The PL intensity of as-obtained CuNCs increased with the concentration of added GSH and, correspondingly, the amount of resultant Cu(I)–GSH precipitate decreased. This implied that the largely excessive GSH was favorable for the “surface etching” route to prepare CuNCs.

Furthermore, the effect of the reaction pH was investigated, and lower pH (pH 2.5) was found to be favorable for the formation of CuNCs (Figure S7). It is noteworthy to mention that the proposed synthetic method exhibits good reproducibility.

To test the generality of the synthetic strategy, we further prepared CuNCs with other thiol ligands, penicillamine (Pen) and cysteine (Cys). As shown in the **Figure 6A**, Pen-mediated CuNCs was turbid, and exhibited an intense red color when irradiated with a UV lamp. Similar to GSH-mediated CuNCs, a strong emission peak at 645 nm and a minor emission peak at 398 nm were observed when excited at 322 nm. Under basic conditions, strong blue emissive CuNCs were obtained by etching with Cys. The excitation and emission maxima were 382 and 478 nm, respectively (Figure 6B). Thus diverse CuNCs with desired surface functionalities can be obtained by tunable ligands and experimental conditions.

3. Conclusion

In summary, fluorescent CuNCs were synthesized via the process of size-focusing surface etching from nonluminescent nanocrystals. The proposed synthetic approach is facile and versatile, offering access to fluorescent CuNCs protected by the desired thiol ligands. Using GSH as a model ligand, the smallest cluster Cu_2 was selectively synthesized to form a nearly monodisperse product, which would allow in-depth studies of the intriguing properties of CuNCs and promote their further practical application. The as-prepared CuNCs emitters are AIEE active, emitting more efficiently while in an aggregated state rather than in a dispersed state. It should be noted that the as-prepared CuNCs are also relatively inexpensive and stable to ambient conditions, making them promising for application in the construction of light-emitting diodes, chemosensors, and bioimaging systems.

4. Experimental Section

Chemicals and Materials: L-ascorbic acid (AA), cysteine (Cys) and glutathione (GSH) were purchased from Sigma-Aldrich (St. Louis, MO, USA). D-penicillamine (Pen) was obtained from Alfa Aesar (Tianjin, China). $Cu(NO_3)_2 \cdot 3H_2O$ and other reagents were purchased from Beijing Chemical Works (Beijing, China). Water used throughout all experiments was purified with a Millipore system (18.2 M Ω). All glassware was washed with aqua regia and rinsed with ultrapure water.

Apparatus and Characterization: Absorption spectra were recorded on a Cary 50 Scan UV-visible spectrophotometer (Varian, USA). Steady-state fluorescence measurements were performed on a Fluoromax-4 spectrofluorometer (Horiba Jobin Yvon Inc. France). The quantum yields were measured directly with the absolute quantum yield measurement system C9920-02 (Hamamatsu Photonics K. K., Japan), which comprises an excitation light source, monochromator, an integrating sphere capable of nitrogen gas flow and a CCD spectrometer for detecting the whole spectral range simultaneously. The luminescence decay curves were measured on a FLS920 spectrofluorometer

(Edinburgh Instruments, UK) equipped with EPL375 pulsed laser diode (ns timescale), and a Lecroy Wave Runner 6100 Digital Oscilloscope (1 GHz) using a tunable laser (pulse width = 4 ns, gate = 50 ns) as the excitation (Continuum Sunlite OPO) (μ s timescale). FTIR spectra of KBr powder-pressed pellets were recorded on a VERTEX FTIR spectrometer (Bruker) scanning from 4000 to 400 cm^{-1} . Transmission electron microscopy (TEM) was performed on JEM-2100F high-resolution transmission electron microscope (Netherlands) operated at 200 kV. The X-ray photoelectron spectroscopy (XPS) measurements were carried out using an ESCALAB-MKII 250 photoelectron spectrometer (VG Co.) with Al $K\alpha$ X-ray radiation (1486.6 eV) for excitation. Matrix-assisted laser-desorption ionization time of flight mass spectrometric (MALDI-TOF MS) studies were conducted using Bruker autoflex III smartbeam MALDI-TOF/TOF-MS (Germany). The spectra were collected in negative-ion mode using the matrix sinapinic acid.

Synthesis of Cu Nanocrystals: Briefly, AA (0.1761 g) was dissolved in water (10 mL) to form a clear solution. Then a $\text{Cu}(\text{NO}_3)_2$ aqueous solution (0.1 M, 1 mL) was added dropwise into the AA solution under vigorous stirring. The reaction mixture was stirred for 1 h at room temperature (20 $^\circ\text{C}$).

Synthesis of GSH-CuNCs: The as-prepared Cu nanocrystals solution (0.3 mL) was added to GSH aqueous solution (0.05 M, 3 mL) and kept for stirring for 5 h at room temperature. As the reaction progressed, white precipitation was generated. The resulting supernate was collected by centrifugation at 8000 rpm for 10 min. The CuNCs were precipitated from the supernate by the addition of ethanol and washed with ethanol (87.5 vol%) repeatedly for three times. The product was freeze-dried and stored in the refrigerator (ca. -20 $^\circ\text{C}$) for long-term preservation.

Synthesis of Pen-CuNCs: The as-prepared Cu nanocrystals solution (0.3 mL) was added to Pen aqueous solution (0.01 M, 3 mL) and kept for stirring for 5 h at room temperature. The resulting precipitate was collected.

Synthesis of Cys-CuNCs: The as-prepared Cu nanocrystals solution (0.3 mL) was added to Cys aqueous solution (0.1 M, 3 mL) containing 0.15 M NaOH and kept for stirring for 5 h at room temperature.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

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